

CS101 - Highlighted Handout

#1 the_handouts_properly

Source: review-driven

Key Definitions:

- **Syntax Errors**: Occur when the code violates language rules.
- **Semantic Errors**: Errors that occur because of incorrect logic or data types in the program.
- **Run-Time Errors**: Errors detected during execution.

Important Concepts:

1. **Review Methods**:
 - Read, understand the problem
 - Get all required data
 - Design and write test cases
 - Write code on paper, hand-check it, type it in, run tests
2. **Testing & Debugging**:
 - Testing: Determine defects.
 - Debugging: Find exact defect locations.
3. **Defect Classification**:
 - Syntax Errors: Violate language rules.
 - Semantic Errors: Logic or data types issues.
 - Run-Time Errors: Occur during execution.
4. **Usability in Web Design**:
 - Success is user-defined, not builder-focused.
 - Use pictures sparingly and motion for attention, not decoration.
5. **PDA (Personal Digital Assistant)**:
 - A small mobile device with computing capabilities.
 - Common products include HP Palmtop and PalmPilot.
 - Features like schedule storage and address book management.

Likely Exam Points:

1. Definitions of Syntax, Semantic, and Run-Time Errors.
2. Review Methods for identifying defects in code.

3. Understanding the importance of testing versus debugging.
4. Classification of errors (Syntax, Semantic, Run-Time).
5. Defining usability success criteria in web design.
6. Key features of PDAs.

Critical Formulas/Examples:

- **No: Get it** and **Else assume it. State it explicitly** are key review steps to ensure all required data is available.
- **Run-time errors** examples like ``a = b + * c`` or ``receiver = reciever + 2`` highlight the importance of checking syntax.

Summary:

This course material emphasizes the importance of reviewing code, testing for defects, and understanding error classifications. Key concepts include syntax, semantic, run-time errors, review methods, usability in web design, and features of PDAs. The definitions and examples provided are crucial for exam preparation.

#2 the_handouts_carefully

Source: review-driven

Key Definitions & Concepts:

1. **Design & Code Reviews**:
 - Humans often see what is supposed to be there instead of what is actually there.
 - Another pair of eyeballs (e.g., reviewers) may not have the same problem, especially if they were not involved in building the design or code.
2. **Review Methods**:
 - **Read, Understand the Problem**: Ensure all required data are available.
 - **Do You Have All the Required Data?**
 - Get it: If not, obtain it.
 - Assume it: State it explicitly if necessary.
 - **Design**: Develop a clear design before coding.
 - **Write Test Cases**: Create test cases to ensure correctness.
 - **Code on Paper**: Write code on paper for initial checks.
 - **Hand-Check & Type in**: Perform final checks and type the code into the system.
 - **Errors? Fix & Redo**: Address errors, fix them, and redo if necessary.
3. **System Architecture**:
 - A collection of elements working together to produce a result not achieved by individual parts alone.
 - Components: Define what makes up the system.

- Connections: Establish how components interact.
- Constraints: Set rules on interactions between components.

Likely Exam Points:

1. **Review Methods and Practices**:

- Understanding different review methods (e.g., reading, understanding, design, test cases).
- Applying these practices to improve code quality and correctness.

2. **System Architecture**:

- Defining the structure of a system.
- Components: Identifying what makes up the system.
- Connections: How components interact.
- Constraints: Rules on how components should work together.

3. **Code Review Process**:

- Steps involved in reviewing code (e.g., reading, understanding, design, test cases).
- Techniques for fixing errors and redoing parts of a project if necessary.

Critical Formulas/Examples:

- **Review Methods**:

- Example: Design a Web page that displays the result of rolling a 6-sided die on user command.
- Example: Combine these words into a sentence i.e., take these strings and concatenate them into one.

- **System Architecture**:

- Example: Define components (e.g., HTML, CSS, JavaScript) and connections between them.
- Example: Constraints: Ensure that components work together in a way that achieves the desired result without redundancy or conflict.

Summary:

The course material emphasizes the importance of thorough code reviews and understanding system architecture. Key points include reading and understanding problems, designing before coding, writing test cases, and performing final checks on paper before typing into the system. Understanding these concepts is crucial for exam preparation as it tests your ability to apply best practices in software development and design.

#3 the_last_lectures_carefully

Source: review-driven

Key Definitions:

- **Algorithm**: A set of well-defined instructions that solve a specific problem or perform a task.
- Example: Sorting algorithms (e.g., Bubble Sort, Quick Sort).

Important Concepts:

1. **Turing Test**:
 - A measure of a machine's ability to mimic human intelligence.
2. **Vacuum Tube**: An early electronic device used in computing.
3. **ABC Language**: A programming language from the 1930s.

Likely Exam Points:

- Definitions and explanations of key concepts like algorithms, Turing Test, vacuum tube.
- Understanding how these relate to modern computing paradigms (e.g., machine learning, artificial intelligence).
- Application examples where these concepts are used in practice.

Critical Formulas/Examples:

- **Algorithm Definition**: A set of steps that solve a problem or perform a task.
 - Example: Sorting algorithm like Bubble Sort or Quick Sort.
- **Turing Test Explanation**: Discussing how it measures machine intelligence and its relevance to modern computing.
 - Example: Comparing the Turing test with current AI capabilities.

Concise Highlighted Summary:

The last lecture focused on algorithms, which are sets of well-defined instructions for solving a problem. Key concepts include:

- **Algorithm Definition**: A set of steps that solve a specific task or problem.
- **Turing Test Explanation**: Measures machine intelligence and relates to modern computing.
- **Vacuum Tube**: Early electronic device used in computing.

Key exam points:

1. Definitions and explanations of key concepts like algorithms, Turing test, vacuum tube.
2. Understanding how these relate to modern computing paradigms.
3. Application examples where these concepts are used in practice.

This summary is likely to appear on exams as it covers the most important definitions and concepts from the course material.